

The G3O Initiative: Building a Global Panel of Government GenAI Use*

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Abstract

This paper introduces G3O (Global Government GenAI Observatory), a cross-national, cross-institutional measurement project designed to track governmental adoption and governance of generative AI over time. We motivate the research agenda, show why existing evidence is insufficient for cumulative research and policy analysis, and outline a human-centered measurement design that produces a versioned panel dataset with transparent validation.

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Rapidly, though unevenly, public institutions around the world are relying on generative AI. New York City introduced a chatbot for constituency service (NYC, 2023); Kenya’s judiciary seeks to integrate GenAI into its procedures (Judiciary, 2025); Albania’s Prime Minister, Edi Rama, even dreams of a ministry run by GenAI, whose algorithms would be immune from corruption and accelerate accession to the EU (Nani, 2025). Although anecdotal examples abound, there is no systematic evidence base beyond limited country-specific case studies (Chen, Gascó-Hernandez and Esteve, 2024; Bright et al., 2025; Liu, Geng and Hart, 2025) or highly aggregated institutional initiatives (OECD, 2025b). As scholars, journalists, policymakers, and citizens consider the relationship between GenAI and public policy, it is vital we understand who is adopting GenAI, for what tasks, and under which safeguards (Pulapaka, Godavarthi and Ding, 2024; Beltran, Ruiz Mondragon and Han, 2024).

Indeed, we stand at a critical juncture. Governments are adopting GenAI at a pace that outstrips experts’ ability to suggest best practices (Bommasani et al., 2025). Our project promises to measure, comprehend, and evaluate the adoption of GenAI at this vital moment, while practices are still forming, letting us document trajectories, not merely outcomes, and provide a frame for comparative learning across jurisdictions. The **Global Government GenAI Observatory (G3O)** addresses this by building a versioned, open-source panel of *institution × quarter* observations that make public-sector GenAI use visible, comparable, and auditable at scale.

G3O will deploy an open-source database of GenAI use across the global universe of public institutions - about 700,000 between national and subnational governments, parliaments, courts, ministries, and agencies (OECD, 2025d). On a quarterly basis, we will run multilingual, institution-specific queries, scrape available web data from top search results, and parse each page with fine-tuned LLM API calls to extract structured fields capturing a host of relevant information, including use case (statements, policies, pilots, initiatives, etc.), tool or vendor, dates, scope, and safeguards. Because many consequential deployments and governance practices are internal and not publicly documented, G3O pairs web-based mea-

surement with a complementary multilingual staff survey to capture internal use, policies, and pilots that would otherwise be invisible. Because many consequential deployments and governance practices are internal and not publicly documented, G3O pairs web-based measurement with a complementary multilingual staff survey to capture internal use, policies, and pilots that would otherwise be invisible.

Our first contribution is empirical. G3O will significantly expand our current understanding of how and where GenAI has been adopted by public institutions, shifting our attention from illustrative cases to population-level observation. This enhanced optic enables comparative inference about diffusion, governance, and outcomes.

The second facet of the broader G3O project is theoretical. We will leverage our unique data to advance a coordinated research agenda on the causes and consequences of GenAI adoption. Our aim is to distribute early data access to research teams and coordinate projects. Bringing classic questions in comparative politics and public administration to bear, this program will ask who is introducing GenAI and why, how does GenAI impact governance and policy outcomes, and reciprocally, how does the adoption of GenAI by governments affect civic participation and trust in institutions. Further avenues of research might include determinants of “cheap talk” (e.g. statements without follow-up) versus actual implementation, as well as the publicity of GenAI initiatives.

A Nascent Research Agenda: Opportunities, Challenges, and the Need for Better Data

A fast-growing academic literature is beginning to analyze generative AI as a consequential development in public administration and digital government, raising foundational questions about state capacity, the organization of bureaucratic work, and citizen-state relations. Across several initial stands of scholarship, researchers are identifying opportunities for improved governance as well as potential pitfalls of rapid and unchecked technological adoption.

Early contributions emphasize that GenAI can reshape the production of core adminis-

trative outputs. Such tasks include drafting, summarizing, and synthesizing large volumes of policy-relevant text, which can reduce bureaucratic workload and accelerate internal workflows (Dunleavy et al., 2006; Pulapaka, Godavarthi and Ding, 2024; Bright et al., 2025; Liu, Geng and Hart, 2025). At the same time, these capabilities introduce new forms of discretion, dependence on vendors, and organizational learning problems. This motivates a first cluster of research questions: when does GenAI augment administrative capacity? Which tasks are most amenable to GenAI automation or assistance? How do agencies adapt routines, training, and performance management as language generation becomes embedded in everyday work? (Moynihan, Herd and Harvey, 2015; Bright et al., 2025).

A second line of work focuses on the relationship between the state and citizens. Many early deployments are public-facing, such as chatbots, or at least accompanied by public statements by officials who may try to signal competency. Therefore, GenAI may directly affect how citizens see the state, from perceived service quality, to transparency and fairness, and ultimately trust and legitimacy. Early work on citizen preferences highlights that design choices for AI-enabled service encounters create several trade-offs between convenience, responsiveness, accountability, and procedural expectations (Hemesath and Tepe, 2024; Moore, 1995). At the same time, scholars argue that generative systems complicate classic accountability relationships by inserting a novel intermediary between public officials and their constituencies (Bovens, 2007; Elliott, P and MacCarthaigh, 2025). In parallel, an emerging governance and legal literature stresses the need for robust guardrails, particularly where GenAI touches administrative justification, rights, and reason-giving. Several studies call for oversight instruments, guidance, and transparency standards in shaping responsible use (Janssen, 2025; Office of Management and Budget, 2024; Algorithmic Transparency Standard, 2026; Pasquale and Malgieri, 2024; Citron and Pasquale, 2014).

A third set of contributions emphasizes the risks of rapid technological adoption. Direct risks include error and hallucination in high-stakes contexts, algorithmic bias, privacy and data protection failures, cybersecurity vulnerabilities, and the erosion of meaningful human

oversight. These risks are not merely technical. They are institutional, because GenAI is often integrated into complex service systems, procured under time pressure, and operated by heterogeneous users with uneven training (Beltran, Ruiz Mondragon and Han, 2024; Janssen, 2025). Indirect risks are equally salient: GenAI may deepen inequality if benefits accrue disproportionately to well-resourced agencies or digitally advantaged citizens; it may amplify surveillance and data extraction pressures; and it may increase administrative burden for marginalized groups when automated interfaces become the default channel for interacting with the state (Moynihan, Herd and Harvey, 2015; Eubanks, 2018; Benjamin, 2019; Pressly, 2024). These concerns reinforce critiques that transparency alone is insufficient, and that meaningful accountability requires novel institutions, careful incentive structures, and auditability, not just disclosure (Ananny and Crawford, 2018; Electronic Privacy Information Center (EPIC), 2023).

Overall, this literature remains in its infancy, both theoretically and empirically. Much of what we know comes from early conceptualizations and normative arguments about governance and rights (Bright et al., 2025; Janssen, 2025; Pasquale and Malgieri, 2024; Beltran, Ruiz Mondragon and Han, 2024). Data-driven studies are scarce and rely on small-N cases, pilots, and surveys in a limited set of settings. Yet the empirical reality is moving faster than scholarship can track: adoption is diffuse, heterogeneous, and often poorly documented, limiting our ability to understand the phenomenology of GenAI in government, compare jurisdictions, test theories of diffusion and capacity building, and evaluate consequences for governance and public trust (Bommasani et al., 2025).

Theorizing GenAI Adoption by Governments

To move from descriptive inventories of deployments to a comparative research program, we require a baseline conceptual model of why public institutions adopt GenAI, how adoption diffuses across jurisdictions, and when adoption translates into meaningful organizational change. Following a long-standing literature on public policy innovation and technological

change, we conceptualize GenAI adoption as a multi-step process, from agenda-setting to exploration and procurement, from piloting to scaling and routinization. Throughout these stages, organizational actors enact a general-purpose technology in context, selecting use-cases, governance arrangements, and integration pathways that are shaped by institutional constraints (Fountain, 2001; Kingdon, 1995; March and Olsen, 1984). This lens is particularly useful for GenAI because early public-sector uptake is characterized by high uncertainty about performance and risk, rapid vendor-driven innovation, and uneven internal capacity to govern, audit, and secure new systems (Janssen, 2025; Beltran, Ruiz Mondragon and Han, 2024; Bommasani et al., 2025).

A simple theoretical decomposition organizes the drivers of adoption into demand, supply, and constraints. On the demand side, public institutions face workload pressures, service backlogs, and fiscal constraints, and they respond to political mandates to modernize or demonstrate responsiveness. These conditions might increase the perceived value of tools that promise faster drafting, summarization, and citizen assistance (Wilson, 1989; Lipsky, 1980; Moynihan, Herd and Harvey, 2015). On the supply side, vendors and “GovTech” entrepreneurs actively market “turn-key” solutions, lowering the fixed costs of adoption and offering procurement pathways that allow rapid experimentation. Supply conditions are further shaped by local data availability, interoperability, and existing digital infrastructure (Fountain, 2001). Finally, adoption is bounded by several types of constraints. First, there are legal and regulatory risks, including administrative law concerns around justification, and due process, as well as privacy and security concerns. In turn, the state’s own capacity to govern the systems it deploys modulates these risks (Hood, Rothstein and Baldwin, 2001; Power, 1997; Pasquale and Malgieri, 2024). Finally, public institutions must consider workforce norms and union constraints, especially when new technologies threaten the job security of public employees, a protected labor category in most contexts. In this framework, adoption is most likely when demand is high and supply is available, conditional on governance capacity that can render experimentation politically and organizationally viable.

This high-level conceptual model immediately implies several families of testable hypotheses that comparative data can adjudicate. First, **diffusion and policy learning**: GenAI practices should diffuse via learning from perceived successes, emulation under uncertainty, competition for investment or reputation, and coercion through higher-level rules or procurement standards (Berry and Berry, 1990; Rogers, 2003; Shipan and Volden, 2008). Institutional pressures may also yield isomorphism: agencies adopt similar GenAI policies and institutional forms (e.g. “responsible AI” guidelines) not only because they are efficient, but because they are legitimating responses to norms and expectations (DiMaggio and Powell, 1983; Tolbert and Zucker, 1983). Second, **bureaucratic politics and principal-agent dynamics**: the distribution of discretion and risk within bureaucracies shapes uptake. Where elected principals seek visible modernization, agencies may prefer low-risk public statements and narrow pilots; where street-level workers experience acute workload pressures, adoption may be driven bottom-up, and the key question becomes whether GenAI shifts discretion upward, through standardized prompts, templates, and monitoring, or downward, through new tools that expand day-to-day autonomy (Lipsky, 1980; Brehm and Gates, 1997; Wilson, 1989). Third, **public legitimacy and contestation**: adoption should be more sustainable when governance is credible. Transparency, appeal pathways, and auditability are likely to mediate citizen acceptance and political backlash, especially for public-facing systems or uses that plausibly affect rights and service access (Bovens, 2007; Power, 1997; Citron and Pasquale, 2014). In short, GenAI adoption is not a simple technocratic choice; it is an institutional choice that balances administrative capacity, technical risk, and political legitimacy.

Why Existing Evidence is Insufficient

Despite the rapid proliferation of public statements, pilots, and deployments, the empirical evidence base on government GenAI adoption remains fragmented, geographically biased, and ill-suited to comparative inference. The most systematic structured inventories are con-

centrated in Europe and the United States. In Europe, the European Commission’s Public Sector Tech Watch (PSTW) provides a large, curated set of emerging-technology adoption cases ([European Commission, Joint Research Centre, 2025a](#); [European Commission, 2025](#)). Specific countries, like the US and the Netherlands, have increasingly been pushed public institutions and agencies toward inventory-style reporting and oversight ([U.S. Government Accountability Office, 2025](#)). Such initiatives offer valuable descriptive insights and anecdotal evidence, but they lack the systematic coverage required for scientific analysis and robust monitoring ([OECD, 2025a](#)).

These limitations matter because many existing inventories share three structural constraints. First, they are frequently snapshots rather than panels, limiting the ability to study trajectories, diffusion dynamics, or reversals. Second, coverage is often incomplete across sectors and levels of government. National executives and flagship digital-government bodies are over-represented, while subnational governments, semi-autonomous agencies, courts, and street-level bureaucracies are systematically under-measured. Third, current coverage is heavily skewed towards Europe and the US.

To make these gaps concrete, we harmonize and present descriptively three complementary sources: OECD case-based documentation, the Policy Innovation Lab’s catalogue, and the US Federal AI Use Case Inventory. Additionally, we collect a convenience sample through structured, AI-augmented web searches ¹. [Figure 1](#) summarizes the resulting global distribution. Our convenience sample strategy recovers use cases by governments across the world, providing illustrative evidence of widespread adoption that is unrepresented in existing sources.

The temporal patterns from this sample underscore the speed of adoption and the need for more frequent and systematic continuous monitoring. [Figure 2](#) shows the monthly, cumulative series for the harmonized data. In just over three years since the release of GPT3.5 (November 2022), we already have more than 1000 documented use cases of the technology,

¹We describe the convenience-sample procedure in [Appendix B](#).

Figure 1: Global distribution of documented government GenAI use cases

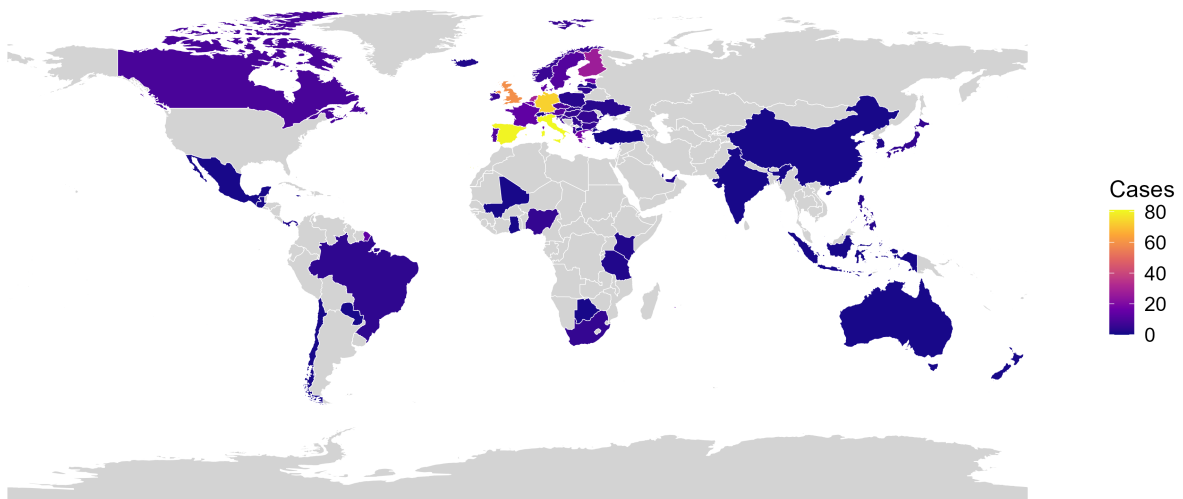
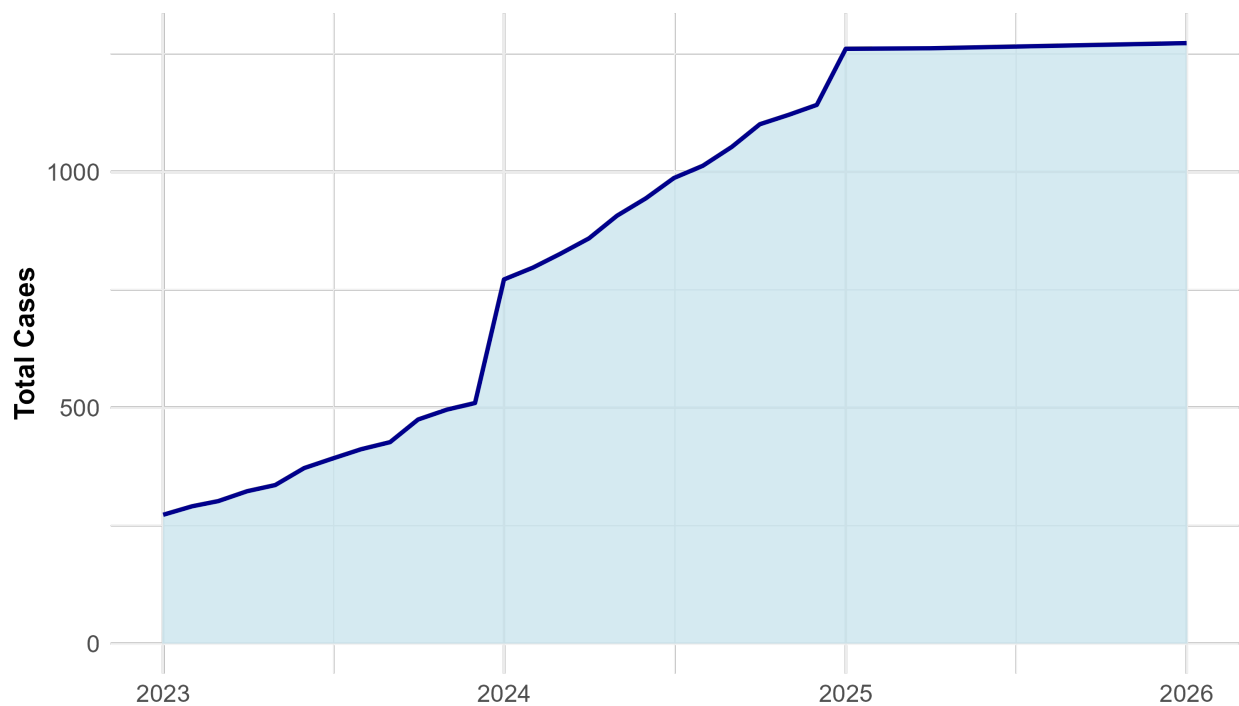
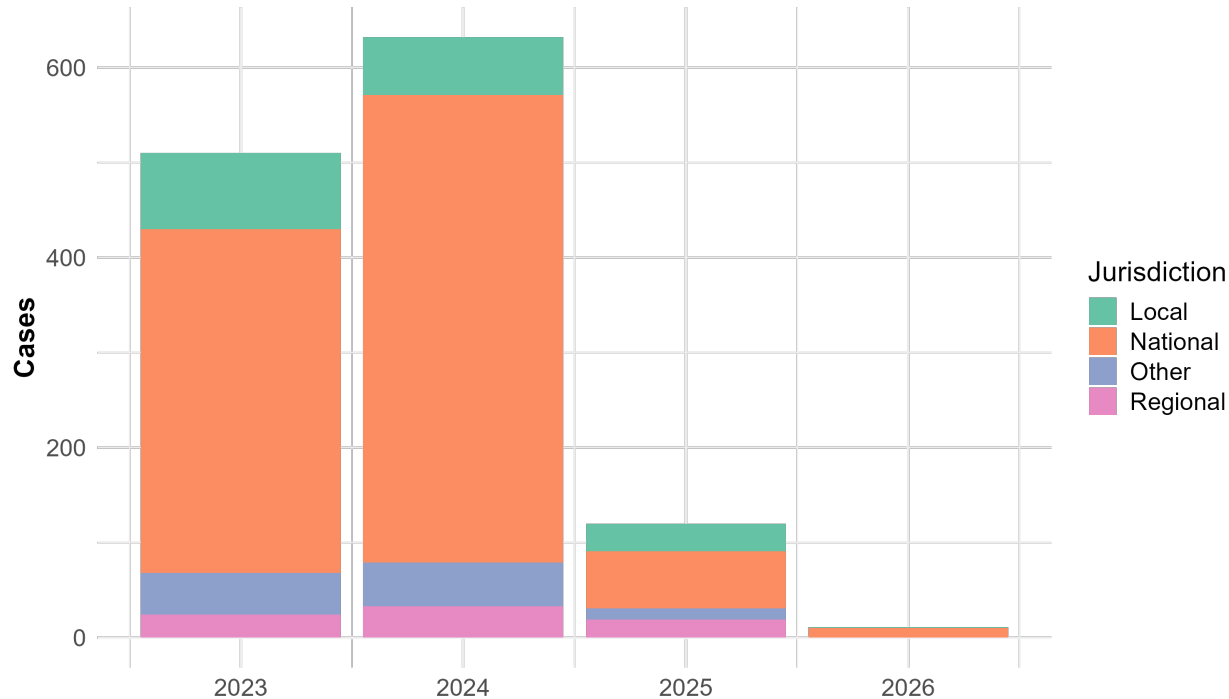


Figure 2: Cumulative Documented Case Count



likely a large underestimate. Given with its potential impact, this rapid pace of change necessitates an equal effort in terms of tracking and analysis. Similarly, [Figure 3](#) displays how adoption is taking place across levels of government, emphasizing the empirical need for a focus on subnational jurisdictions.

Figure 3: Distribution of GenAI Uses by Government Type



Documented cases by year/source component.

Introducing the Global Government GenAI Observatory (G3O)

The Global Government GenAI Observatory is a public, auditable dataset that measures GenAI activity across government institutions using official and other high-credibility public sources. It is built to address the limitations of existing inventories, especially uneven global coverage and limited institutional granularity. G3O is organized around repeated measurement, record-level provenance, and versioned releases.

G3O adapts the Public Sector Tech Watch (PSTW) case structure to a global, versioned

institution $\times$ *quarter* panel, and extends it to GenAI-relevant adoption stages and governance fields (European Commission, Joint Research Centre, 2025b; Tangi et al., 2023). We complement this with risk-oriented governance fields aligned with OECD guidance for public-sector AI (OECD, 2025d).

Each institution-by-time observation includes five field families. Institution metadata captures country, level of government, branch or sector, functional domain, and a persistent identifier for longitudinal linkage (European Commission, Joint Research Centre, 2025b). GenAI activity fields capture activity type, adoption stage, access type, interaction type, and scope notes (Tangi et al., 2023). Implementation fields capture tool or vendor when disclosed, deployment mode, and integration into workflows, with additional operational notes only when supported by sources (OECD, 2025d). Governance fields capture guardrails and oversight such as human review, transparency and notice, privacy and security (OECD, 2025d). Outcome fields capture performance and incident indicators only when explicitly documented.

Table 1 summarizes the activity typology and the operational dimensions coded for each category. The typology follows the PSTW logic of structured representation while extending measurement to adoption stage, integration, and GenAI-relevant guardrails (European Commission, Joint Research Centre, 2025b; Tangi et al., 2023; OECD, 2025d).

G3O anchors discovery and measurement in a structured universe of public institutions spanning national and subnational government across all branches of government and main national-level bureaucratic agencies. Table 2 summarizes the current sampling frame. We assemble the universe relying on comprehensive registries. For subnational government, we rely on the OECD/UCLG World Observatory on Subnational Government Finance and Investment, which reports territorial organization and unit counts by tier across countries (OECD, 2022, 2025c). For national-level institutions, we combine cabinet and ministry portfolios from WhoGov, parliamentary chamber lists from IPU PARLINE, and the Judicial branch field from the CIA World Factbook to enumerate national judicial bodies (Nyrup,

Activity type	Operational definition and typical evidence	Key operational dimensions coded
Policy and guidance	Executive orders; agency memos; acceptable-use rules; procurement standards; internal guidance on permissible tools and data handling. Evidence typically comes from official policy documents or government portals.	Stage; scope; internal versus citizen-facing relevance; guardrails stated.
Pilots and experiments	Limited trials, sandboxes, proof-of-concept deployments, or time-bounded experiments. Evidence typically includes pilot announcements, procurement or tender records, or program pages naming the participating public body.	Stage; access; integration; tool or vendor disclosure; evaluation claims; uncertainty flags.
Programs and initiatives	Cross-agency programs, governance bodies, training programs, or platform rollouts spanning multiple units. Evidence often includes program charters, budget lines, or cross-agency announcements.	Lead unit; coverage; governance structures; procurement pathway; workforce enablement.
Internal operational use	Routine internal use for drafting, summarization, knowledge search assistants, translation, internal helpdesks, or developer tooling. Evidence may appear in approved-tool lists, internal-tool announcements, or procurement and deployment notes.	Use-case tags; integration; affected workflow; guardrails; outcome claims when reported.
Public-facing services	Citizen- or business-facing chatbots; service navigation; translation; complaint handling; eligibility or case triage support. Evidence typically includes live service portals, official releases, or procurement tied to a named service channel.	Interaction type; channel; safeguards and escalation; accessibility claims; incidents when documented.

Table 1: G3O measurement typology and operational dimensions. The typology follows PSTW-style structured representation while extending measurement to adoption stage, integration, and GenAI-relevant guardrails for longitudinal and cross-institutional comparison (European Commission, Joint Research Centre, 2025*b*; Tangi et al., 2023; OECD, 2025*d*).

Bramwell and Lönn, 2024; Inter-Parliamentary Union, 2025; Central Intelligence Agency, 2026). Finally, for bureaucratic agencies, we rely on WikiData’s GovDirectory.

Table 2: Institutional universe for G3O.

Unit type	Source / code	Total units	Countries with data
Municipal	OECD SNG-WOFI – MUN	629,328	133
Intermediate	OECD SNG-WOFI – INT	9,986	40
Regional	OECD SNG-WOFI – REG	2,433	100
Ministries	WhoGov	2,784	131
Legislative bodies	IPU PARLINE	196	132
Judicial bodies	CIA World Factbook	1,768	133
Government agencies	WikiData, GovDirectory and national directories	~28,800	133
Grand total		675,295	133

The baseline universe combines globally scoped registries for subnational tiers with national-level sources for ministries, legislatures, and judicial bodies. The agency layer is under construction. Sources: OECD/UCLG SNG-WOFI (OECD, 2022, 2025c), WhoGov (Nyrup, Bramwell and Lönn, 2024), IPU PARLINE (Inter-Parliamentary Union, 2025), CIA World Factbook (Central Intelligence Agency, 2026).

Data Collection Pipeline

G3O follows a three-layer pipeline deployed at the level of individual institution: discovery, extraction, and cross-validation. Discovery identifies high-credibility sources documenting GenAI activity for the specific institution using unit-specific systematic retrieval. Retrieval relies on automated searches through the google search API. Queries combine standardized institution identifiers with GenAI terms, common tool names, and task descriptors, and are executed across all relevant official languages. For each candidate source, we record URLs, retrieval dates, and publication or issuance dates when available (Tangi, Combetto and Martin Bosch, 2024; European Commission, Joint Research Centre, 2025b). Discovery also identifies official institutional contact emails and, when publicly listed, technical-unit contacts (e.g., IT, digital service, CIO/CTO, security) used to distribute the voluntary multilingual survey.

Extraction converts retrieved sources into structured records using schema-first parsing.

HTML and PDF materials are mapped into fixed fields with explicit coding conventions, and records retain short evidence snippets. This procedure employs LLM API calls following recent methodological guidelines in using GenAI for information extraction from unstructured text (Paci et al., 2025).

Cross-validation entails merging all candidate records. Information about GenAI use is de-duplicated and consolidated using conservative merge rules anchored on institution and distinct activity. Uncertainty flags are used when sources leave stage, ownership, or GenAI status ambiguous. These uncertainty flags support downstream sensitivity analyses and targeted review (Tangi, Combetto and Martin Bosch, 2024). When disagreement between sources arises, the merging process prioritizes primary sources such as government domains, procurement and tender portals, council and parliamentary records, annual reports, and regulator publications, using reputable secondary reporting only when primary documentation is unavailable.

Survey Component: Multilingual Staff Survey

Public web sources likely systematically under-measure internal GenAI use, informal governance, and pilots that are not publicly described. To complement the web-based panel, G3O includes a short, voluntary staff survey that captures private, internal information on GenAI usage and governance. The survey is designed to recover internal-use signals that rarely appear in public documentation. At the same time, it will elicit links or references that help the pipeline discover and validate specific initiatives that are otherwise difficult to retrieve.

Contact collection is integrated into the same institution-level discovery pipeline. Alongside GenAI-relevant sources, the retrieval layer extracts official institutional contact addresses and, when publicly available, emails associated with technical functions. These contacts will then be used to distribute an institution-specific survey invitation tied to the institution identifier and the current quarter.

Survey invitations and the instrument are distributed in the institution’s official language(s), using the same language roster used for multilingual web retrieval. The questionnaire focuses on three aspects of GenAI use: first, whether the institution has formal GenAI policies or guidance and whether norms are consistent when formal policy is absent; second, the prevalence of internal GenAI use and the tools/vendors in use; third, the presence of pilots/initiatives and whether pilots are evaluated against defined success metrics. An open-ended item invites respondents to share links or references to specific pilots/initiatives. The full draft instrument is provided in ??.

Survey responses complement public records rather than replace them. We treat self-reports as additional measurement of internal use and governance where public visibility is limited and targeted discovery inputs: when respondents provide links or specific initiative names, these are ingested as candidate sources and processed through the same extraction and cross-validation workflow as other retrieved materials. This design both augments coverage of otherwise non-public internal adoption and helps fill gaps in the recovery of specific use cases, while preserving the provenance and audit logic of the core dataset.

Validation

To validate discovery and extraction procedures, Each wave includes human review on a stratified sample drawn across regions, languages, institution types, and activity types, further blocked on uncertainty estimates. Manual collection follows the same inclusion rule and schema as the automated pipeline, producing a comparison set used to estimate error rates and characterize performance by region, language, and institution type.

Validation proceeds in two steps targeting different failure points. First, discovery validation tests whether the pipeline recovers all relevant sources for a given institution. Second, extraction validation tests whether coded fields accurately reflect the retrieved sources (Tangi, Combetto and Martin Bosch, 2024). A random subset of records is double-coded, with disagreements adjudicated using a third round of manual verification, to estimate in-

tercoder and intracoder error rates.

Limitations

Several constraints remain even with repeat measurement and validation. First, G3O is limited by public visibility. Many internal or sensitive deployments are not disclosed, and some procurement and security-relevant information is withheld. The survey arm can reduce this gap, but nonresponse, uneven reach to technical staff, and strategic or mistaken reporting constrain the completeness and reliability of what it can recover.

Second, large-scale web collection is bounded by legal and practical access constraints. Robots exclusions, site terms, rate limits, authentication barriers, and inaccessible procurement portals can prevent retrieval for some institutions and jurisdictions.

Third, longitudinal comparability is constrained by web instability. Pages move, links rot, and content is revised without notice, which can create apparent change unrelated to adoption dynamics.

Fourth, governance and impact measures are necessarily thin. Guardrails are often observable only as stated commitments, and their presence does not imply enforcement. Similarly, outcome indicators are selectively reported and rarely comparable across jurisdictions, so G3O measures institutional activity and stated governance more reliably than realized performance or harm.

Finally, the institutional sampling frame is incomplete and evolves over time. Global registries lag administrative reforms, and cross-national enumeration of agencies and sectoral public bodies requires country-specific sources and harmonization rules. Early releases may therefore over-represent institution types with strong registries relative to specialized agencies, with coverage reported transparently as the frame expands.

Conclusion and Plans for the Future

G3O is designed as measurement infrastructure for a rapidly evolving phenomenon: it provides a stable institutional sampling frame, repeated observation on a fixed cadence, auditable provenance for each record, and a complementary survey arm to surface internal use and governance practices that are not publicly disclosed. The project’s long-run contribution, however, depends on whether this infrastructure catalyzes cumulative, comparative research and practical learning among governments and other stakeholders. We therefore plan two complementary extensions: a coordinated research program and a policy-facing dissemination and feedback strategy.

A central goal of G3O is to lower the transaction costs of cumulative learning. We take inspiration from coordinated research models that emphasize common measurement, shared standards, and integrated synthesis across otherwise independent teams ([Leaver, 2019](#); [Evidence in Governance and Politics \(EGAP\), 2021](#)). Concretely, we will treat the G3O database as a research public good that enables coordinated projects on adoption, diffusion, guardrails, and outcomes. This coordination will take three steps. First, we will circulate a call for papers to gather an initial round of interested research teams, each focusing on a different aspect or context using the G3O data yet employing a common set of theoretical and empirical approaches. These teams will receive early access to the data collected in the first wave of G3O data collection. Second, we will convene a research conference for teams to present their findings, provide each other feedback, and further coordinate on the shared research agenda. Finally, we will seek common publication either as a special issue or an edited volume.

A parallel objective is to support evidence-based governance, especially as public institutions adopt GenAI faster than best practices can consolidate. G3O will publish practitioner briefs and dashboards tailored to policymakers, civil society, and journalists, emphasizing interpretable aggregates (adoption rates, types of use, guardrails) and traceability to sources. Where feasible, we will maintain feedback loops with agencies and public bodies to correct

records, add contextual detail, and improve documentation quality. Over time, the database can also support more systematic monitoring and guidance by identifying which governance and guardrail patterns correlate with fewer reported incidents, more credible accountability mechanisms, or improved service delivery indicators.

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Supplemental Materials for
The G3O Initiative:
Building a Global Panel of Government GenAI

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A Country Sample Selection

This appendix documents the country sampling strategy used for the initial round of the Global Government GenAI Observatory (G3O). The core design objective for Wave 1 is to prioritize *measurable comparability* and *institutional coverage* for foundational units of public administration (subnational governments, ministries, legislatures, and apex judicial bodies), while maintaining broad global reach. We therefore anchor the initial country universe to the set of countries covered by the OECD Subnational Government (SNG) institutional statistics used in our measurement design. This choice yields a large and diverse cross-section of countries for which subnational administrative structure is systematically measured, enabling consistent downstream linkage across institutional layers (e.g., ministries and legislative chambers) and facilitating validation and auditing.

A.1 OECD-based initial country universe

Our initial Wave 1 country universe consists of all unique country codes appearing in the OECD SNG source file (ISO3 codes as provided in the dataset). In the version of the OECD file used for Wave 1, this universe comprises $N = 134$ country codes.²

OECD Wave 1 country list (ISO3 codes).

AGO, ALB, ARG, ARM, AUS, AUT, AZE, BDI, BEL, BEN, BFA, BGD, BGR, BHR, BIH, BLR, BOL, BRA, BRN, BTN, BWA, CAF, CAN, CHE, CHL, CHN, CIV, CMR, COD, COG, COL, CRI, CUB, CYP, CZE, DEU, DJI, DNK, DOM, DZA, ECU, EGY, ERI, ESP, EST, ETH, FIN, FJI, FRA, GAB, GBR, GEO, GHA, GIN, GMB, GNB, GRC, GTM, HKG, HND, HRV, HTI, HUN, IDN, IND, IRL, IRN, IRQ, ISL, ISR, ITA, JAM, JOR, JPN, KAZ, KEN, KGZ, KHM, KOR, KWT, LAO, LBN, LBR, LKA, LSO, LTU, LUX, LVA, MAR, MDA, MDG, MEX, MKD, MLI, MMR, MNE, MNG, MOZ, MRT, MUS, MWI, MYS, NAM, NER, NGA, NIC, NLD, NOR, NPL, NZL, OMN, PAK, PAN, PER, PHL, PNG, POL, PRT, PRY, QAT, ROU, RUS, RWA, SAU, SDN, SEN, SGP, SLV, SOM, SRB, SUR, SVK, SVN, SWE, SWZ, SYR, TCD, TGO, THA, TJK, TKM, TUN, TUR, TZA, UGA, UKR, URY, USA, UZB, VEN, VNM, XKV, ZAF, ZMB, ZWE

A.2 Countries not covered by the OECD Wave 1 universe

To contextualize coverage, we compare the OECD Wave 1 universe to the ISO 3166-1 alpha-3 list (which includes sovereign states as well as non-sovereign territories and special jurisdictions). Relative to ISO 3166-1, the OECD-based universe omits 116 ISO3 codes. These omissions are substantively patterned: they include (i) many *small island states* (particularly in the Caribbean and the Pacific), (ii) a substantial set of *non-sovereign territories and dependencies* (which are in ISO 3166-1 but are not consistently measured as separate

²The OECD file includes the code XKV for Kosovo, which is commonly used in international datasets but is not an official ISO 3166-1 alpha-3 code. We retain XKV for internal consistency with OECD coverage and to preserve joinability across sources.

governmental systems in cross-national administrative datasets), and (iii) a smaller number of *sovereign states* that are not covered in the OECD SNG source file used for Wave 1.

This “OECD-first” strategy is motivated by measurement: Wave 1 aims to establish a high-quality, joinable backbone across institutional layers. The missing set is a natural target for subsequent waves, where we will expand coverage through targeted backfills using additional administrative structure sources and country-specific validation protocols.

Countries missing from the OECD Wave 1 universe (ISO 3166-1 alpha-3).

ABW, AFG, AIA, AND, ASM, ATA, ATF, ATG, BHS, BLM, BLZ, BMU, BRB, BRB, BRB, BVT, CCK, COK, COM, CPV, CUW, CYM, DMA, FLK, FRO, FSM, GGY, GIB, GLP, GRD, GRL, GUM, GUY, HMD, IMN, IOT, IRL, ISL, JEY, KIR, KNA, LCA, LIE, MAC, MAF, MCO, MDV, MHL, MLT, MNP, MSR, MTQ, NCL, NFK, NIU, NRU, PCN, PLW, PRI, PYF, REU, SJM, SLB, SPM, STP, SXM, TCA, TKL, TLS, TON, TTO, TUV, UMI, VCT, VGB, VIR, VUT, WLF, WSM, YEM

Implementation note. The lists above are provided as ISO3 codes for compactness and to facilitate reproducible merges across sources. In our public release, we will provide a cross-walk to standardized country names and to alternative country coding schemes as needed (e.g., ISO2). For Wave 1 internal processing, the OECD-based list constitutes the sampling frame for country-level institution mapping and GenAI use-case discovery; the missing list serves as the pre-specified expansion frame for future waves.

B Convenience Sample for Illustrative Data Collection

We conducted a structured, two-stage web search to identify empirically documented uses of generative AI (GenAI) by government entities in the target region (excluding the United States and EU member states). First, we ran broad discovery queries combining country/agency names with GenAI terms (e.g., “generative AI,” “LLM,” “ChatGPT,” “assistant,” “chatbot,” “drafting,” “summarization,” “translation,” “knowledge base/RAG,” “copilot”), then snowballed from promising leads by following named programs, vendor announcements tied to specific public bodies, procurement/tender records, and official press releases. We prioritized primary sources (government domains and official procurement portals) and used reputable secondary reporting only when primary documentation was unavailable. Each candidate was included only when at least one source explicitly confirmed (1) the government entity, (2) the generative/LLM nature of the system, and (3) the use context; we then extracted a short evidence snippet, de-duplicated to one row per (government entity $\times$ distinct use-case), and coded the resulting cases to the PSTW taxonomy fields, leaving fields blank where unsupported and assigning confidence (High/Med/Low) based on source quality and specificity.

C Survey Instrument

Table A1: Short voluntary survey instrument for technical staff at public institutions (Institution X).

Category	Question	Items (answer values)
Response scales used in this instrument		
Scales	Reusable scales referenced below	<i>Frequency (5-cat)</i> : Daily; Weekly; Monthly; Yearly or less; Never. <i>Agreement (7-pt)</i> : 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither; 5 Somewhat agree; 6 Agree; 7 Strongly agree. <i>Percent ranges</i> : 0%; 1–10%; 11–25%; 26–50%; 51–75%; 76–90%; 91–100%.
A. Respondent	A1. Are you currently working for [Institution X] (employee or embedded contractor supporting core systems)?	Yes / No (terminate if No).
A. Respondent	A2. Which best describes your work?	Technical (IT, engineering, data, security, architecture); Operational/program (non-technical); Leadership/management; Other.
A. Respondent	A3. Tenure at [Institution X]	<6 months; 6–12 months; 1–2 years; 3–5 years; 6–10 years; 11+ years.
B. Internal use	B1. Estimate the share of employees at [Institution X] who use GenAI for work at each level. (One response per row.)	Rows: Daily; Weekly; Monthly; Yearly or less; Never. Columns/answers: Percent ranges (0% ... 91–100%).
B. Internal use	B2. Which GenAI tools are used by employees at [Institution X] for work? (Select all that apply.)	Microsoft Copilot (M365 / Copilot Studio); OpenAI (ChatGPT Enterprise / API-based internal tools); Google (Gemini / Workspace); Anthropic (Claude); AWS (e.g., Bedrock-based tools); Azure OpenAI-based internal tools; Other vendor tool (free text); Institution-built/self-hosted model/tool (including open-source models).

Continued on next page

Table A1 (continued)

Category	Question	Items (answer values)
B. Internal use	B3. For each task below, how often are employees using GenAI for this purpose?	Answer values: Frequency (Daily; Weekly; Monthly; Yearly or less; Never). Task rows: (1) Drafting/editing text (emails, memos, reports); (2) Summarizing documents; (3) Translation/rewriting for clarity; (4) Coding assistance (generate code, refactor); (5) Debugging/explaining code; (6) Data analysis support (e.g., generating queries, interpreting results); (7) Knowledge/FAQ assistance (internal knowledge base, RAG-like assistants); (8) Citizen-facing content drafting (web copy, FAQs, scripts).
B. Internal use	B4. “Overall, employees at [Institution X] are positive about using GenAI for internal tasks.”	Agreement (7-point Likert).
B. Internal use	B5. “Overall, employees at [Institution X] are positive about using GenAI for external/citizen-facing services.”	Agreement (7-point Likert).
C. Policies	C1.1 Does [Institution X] have a formal policy or guidance governing GenAI use by staff?	Yes / No.
C. Policies	C1.5 (Asked only if C1.1 = No) “Even without a formal policy, informal norms and manager guidance about GenAI are consistent across teams.”	Agreement (7-point Likert).

Continued on next page

Table A1 (continued)

Category	Question	Items (answer values)
C. Policies	C2 (Asked only if C1.1 = Yes) Which of the following are in place at [Institution X] ?	Answer values for each statement: Yes / No. Statements: (1) Official list of approved GenAI tools/vendors for work use; (2) Defined approval process for new tools or new use-cases; (3) Clear rules on what data can/cannot be entered into GenAI tools; (4) Required training or onboarding guidance for staff on GenAI use; (5) GenAI use is logged/monitored for security, audit, or incident response (as appropriate); (6) Sensitive/regulated uses require additional review before deployment (e.g., security/privacy/legal).
D. Pilots	D1. In the last 12 months, has [Institution X] run any GenAI pilots or initiatives (internal and/or citizen-facing)?	Yes / No.
D. Pilots	D2. “When [Institution X] runs GenAI pilots/initiatives, they are evaluated against defined success metrics (e.g., quality, cost, time, risk) before scaling.”	Agreement (7-point Likert).
D. Pilots	D3. Please share links or references to any specific GenAI pilots/initiatives you are aware of.	Open text (respondents may paste URLs or name the initiative and where it is described).